Demo: MarketIntel MCP Server

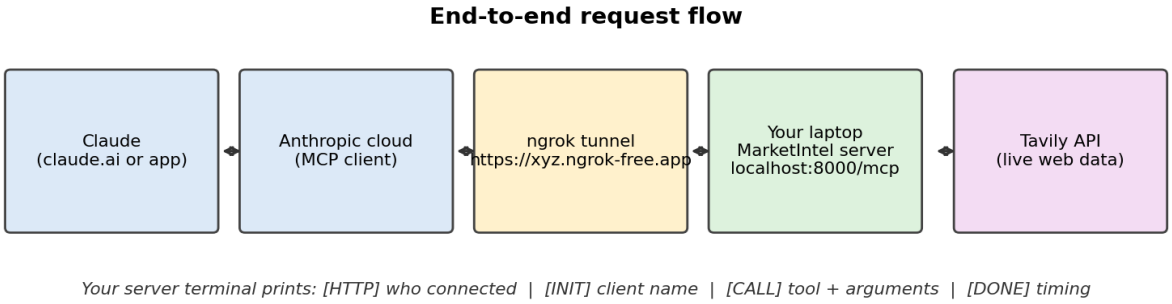
Connect a custom market-research MCP server running on your laptop to Claude

Objective: Build and run a custom MCP server locally, expose it to the internet through a secure tunnel, connect it to Claude as a custom connector, and show live market-research results while the server terminal logs who called what.

Time needed: about 30 to 40 minutes the first time, about 5 minutes once everything is installed.

1. How it works

Claude cannot reach a program on your laptop directly. Custom connectors are called from Anthropic's cloud, so the server must have a public HTTPS address. A tunnel (ngrok) gives your local port that address.



- **Tools** are what Claude calls: company overview, competitors, pricing, news, and an all-in-one brief.
- **Logs** are printed by the server for every hit: raw HTTP request, MCP session start, tool name, arguments, caller IP, and timing.
- **Demo mode:** if you do not set a Tavily key, the server returns built-in sample data for NVIDIA, AMD and Intel, so the demo still works.

Note: This demo is driven by tools. Claude can also surface a server's prompts and resources, but the demo only needs tools, which keeps it simple and reliable.

2. What you need

Item	Why	Cost / notes
Claude account (free works)	Lets you add a custom MCP connector	Free plan can add one custom connector. Pro and Max also work. On Team or Enterprise an Owner must add it first. Usage limits apply.

Item	Why	Cost / notes
Python 3.10+ and uv	Runs the server and manages packages	Free. uv can install Python for you.
ngrok account + authtoken	Public HTTPS URL for your local server	Free account is enough. You get the authtoken from the ngrok dashboard (Step 6). Alternative: Cloudflare Tunnel.
Node.js (optional)	Only needed for the MCP Inspector in Step 5	Skip it and use the fastmcp list / call commands instead.
Tavily API key (optional)	Live web search results	Free tier available. Skip it to use demo data.
The project files	server.py, test_client.py, .env.example	Provided with this guide. Use the pyproject.toml that uv init creates, not another one.

3. Step-by-step setup

Step 1: Install uv

macOS or Linux:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Windows (PowerShell):

```
powershell -c "irm https://astral.sh/uv/install.ps1 | iex"
```

Close and reopen the terminal, then confirm:

```
uv --version
```

Step 2: Create the project

```
uv init marketintel_demo
cd marketintel_demo
uv add fastmcp tavily-python
```

Delete the generated `main.py` and copy these files into the folder: `server.py`, `test_client.py`, `.env.example`.

Folder structure:

```
marketintel_demo/
  server.py          <- the MCP server (tools + logging)
  test_client.py     <- quick local test
  .env               <- your settings (you create this)
  pyproject.toml
```

Step 3: Add your settings (.env)

Copy `.env.example` to `.env`. Paste your Tavily key if you have one; otherwise leave it empty.

```
TAVILY_API_KEY=tvly-xxxxxxxxxxxxxxx
TAVILY_DEPTH=basic
PORT=8000
```

- **Get a key:** sign up at tavily.com, open the dashboard, click the + beside API Keys, name it, and copy it.
- **Credits:** basic search costs fewer credits than advanced. Keep it on basic for the demo.

Never commit .env to Git and never paste the key into server.py. The server reads it from the environment.

Step 4: Run the server

```
uv run server.py
```

You should see the banner. The Mode line tells you whether you are on live data or demo data:

```
======================================================================
 MarketIntel MCP Server
 Mode      : LIVE (Tavily)          <- or DEMO (canned data ...)
 Listening : http://0.0.0.0:8000/mcp     health: http://localhost:8000/health
======================================================================
```

Leave this terminal open. It is your live log window during the demo.

Step 5: Test it locally before going public

Open a second terminal in the same folder:

```
uv run python test_client.py
curl http://localhost:8000/health
```

test_client.py lists the tools and calls three of them. Switch back to the server terminal and confirm you see [INIT], [LIST], [CALL] and [DONE] lines. If you do, the server is ready.

Optional: try the tools by hand. Pick one of these two ways.

Option A: MCP Inspector (visual, needs Node.js). Check Node.js with `node --version` (install on a Mac with `brew install node`). Then run:

```
uv run fastmcp dev inspector server.py
```

- The terminal prints a link to the Inspector, and it includes a security token. Open that full link in your browser.
- Click Connect, open the Tools tab, choose a tool such as list_competitors, fill in the inputs and run it.
- Press Ctrl+C in the terminal to stop it.

Option B: command line (no Node.js needed).

```
uv run fastmcp list server.py
uv run fastmcp call server.py list_competitors name=NVIDIA
```

The first command lists every tool. The second runs one tool and prints its result as JSON, so you can check the output without any AI app.

Step 6: Create an ngrok account and get your authtoken

ngrok will not start a tunnel until it knows which account it belongs to. That link is the authtoken, a long secret string you copy once from your dashboard.

1. Go to **ngrok.com** and click **Sign up**. It is free. Use your email, Google or GitHub, and confirm your email if ngrok asks.
2. After you sign in you land on the ngrok dashboard (`dashboard.ngrok.com`).
3. Open the authtoken page: `https://dashboard.ngrok.com/get-started/your-authtoken`. In the dashboard menu this is under Getting Started > Your Authtoken.
4. Click the copy icon next to the token. Keep it private, like a password. You will paste it into the terminal in the next step.

Step 7: Install ngrok and start the tunnel

1. Install ngrok. On a Mac: `brew install ngrok`. On Windows or Linux, download it from ngrok.com/download.
2. Register your token once (replace the placeholder with the token you copied):

```
ngrok config add-authtoken <YOUR_TOKEN>
```

You should see a message that the authtoken was saved to the configuration file. You only do this once per computer.

3. Make sure the MarketIntel server from Step 4 is still running, then start the tunnel in a third terminal:

```
ngrok http 8000
```

4. Find the **Forwarding** line in the ngrok window and copy the https address, for example `https://a1b2-203-0-113-9.ngrok-free.app`. Your MCP URL is that address plus **/mcp**:

```
https://a1b2-203-0-113-9.ngrok-free.app/mcp
```

5. Verify from a browser: open the address plus `/health`. You should see JSON with `"status": "ok"`. If ngrok shows a warning page first, click Visit Site; that page is only for browsers. The server terminal logs this hit with your IP.

Note: Depending on your ngrok account, the address may stay the same between restarts or change each time. Always copy the Forwarding line again after restarting ngrok. If it changed, edit the URL of the connector in Claude. Cloudflare Tunnel is a free alternative: cloudflared tunnel --url http://localhost:8000

Error ERR_NGROK_4018 or "authentication failed"? The authtoken is missing. Run the add-authtoken command from this step again with the token from your dashboard.

Step 8: Connect Claude

You can use a free Claude account. On the Free plan you can add one custom connector, so if you already added one, remove it first. Menu names change from time to time; the paths below are the current ones from Claude's help pages.

1. Open **claude.ai** in your browser (or the Claude desktop app) and sign in.

2. Go to **Customize > Connectors**. Direct link: `claude.ai/customize/connectors`. In some older versions this is called Settings > Connectors.
3. Click **Add custom connector**.
4. Fill in the dialog:
   - **Name:** MarketIntel
   - **MCP server URL:** your ngrok address ending in /mcp
   - **Authentication:** choose No sign-in (this demo server has no login). If your dialog shows OAuth Client ID and Secret under Advanced settings, leave them empty.
   - **Transport:** leave the default. A URL ending in /mcp is correct.
5. Click **Add**. Watch your server terminal: you should see `[HTTP]` lines and usually `[INIT]` and `[LIST]` as Claude connects and reads the tool list.
6. Open a new chat, click the **+** button, choose **Connectors**, and switch MarketIntel on for this conversation.

Team or Enterprise plan: an Owner must first add the connector in Organization settings > Connectors. You then go to Customize > Connectors, find MarketIntel (it has a Custom label) and click Connect.

4. Running the demo

Arrange the screen so the audience sees Claude on one side and the server terminal on the other.

#	What you type in Claude	What the audience sees in the terminal
1	What tools does the MarketIntel connector give you? List them.	[LIST] tools returned. Shows Claude discovered the server.
2	Use MarketIntel to give me a company overview of NVIDIA.	[CALL] company_overview with its arguments, then [TAVILY] or [DEMO], then [DONE] with timing.
3	Who are AMD's main competitors in GPUs? Use MarketIntel.	[CALL] list_competitors, args name=AMD, category=GPUs.
4	Create a market research report comparing NVIDIA and AMD using MarketIntel. Cover overview, product portfolio, pricing, competitors, recent news, then SWOT and Five Forces. Cite sources.	Several [CALL] lines in a row. Claude chains tools on its own. The main talking point of the demo.
5	Call server_status and tell me how many requests you have made.	[CALL] server_status. The counters match what the audience watched in the log.

Talking points: the model decides which tool to call; the arguments come from the conversation; the server returns structured data; Claude writes the analysis. Point at the log each time to prove the call really hit your machine.

Tip: Claude asks for permission before it uses a tool. Choose Allow once, or Always allow for the demo chat only. Free-plan usage limits can cut a long multi-tool report short, so run prompt 4 once before the audience arrives.

5. Reading the logs

Every line starts with a timestamp and a tag. This is an illustrative example. Your IP, host and client name will differ, and Claude's calls come from Anthropic's servers, not from your browser:

```
[09:21:44.141] [HTTP  ] --> POST /mcp session=- ip=203.0.113.9 host=a1b2.ngrok-free.app
ua='<sent by Claude>'
[09:21:44.141] [INIT  ] New MCP session: client='<name sent by Claude>'
ip=203.0.113.9 ...
[09:21:45.112] [LIST  ] Client asked for tools -> returned 7: ['company_overview', ...]
[09:21:47.119] [CALL  ] >>> TOOL 'competitor_brief' args={"company": "AMD", "category":
"GPUs"}
[09:21:47.119] [CALL  ]     caller: ip=203.0.113.9 host=a1b2.ngrok-free.app ua='<sent
by Claude>'
[09:21:47.121] [DEMO  ] serving canned data for query='Company overview of AMD ...'
[09:21:47.121] [DONE  ] <<< TOOL 'competitor_brief' OK in 2 ms, response ~1672 chars
```

Tag	Meaning
HTTP	Raw request reaching the server: method, path, MCP session id, caller IP (from the ngrok X-Forwarded-For header), Host and User-Agent. Also prints the status code and time on the way out.
INIT	A new MCP session started, with the client name and version that Claude reports.
LIST	The client asked which tools exist.
CALL / DONE / ERROR	A tool started with its arguments, finished with timing and response size, or failed with the error.
TAVILY / DEMO	Whether the data came from a live Tavily search (with the query and result count) or from the built-in sample data.
HEALTH	Someone opened /health.

A request to /health is also a quick way to see a caller IP: Claude's calls come from Anthropic's servers, while your own browser test shows your IP.

6. Troubleshooting

Symptom	Fix
No Add custom connector option in Claude	Open Customize > Connectors (older versions: Settings > Connectors). On Team or Enterprise an Owner must add the connector in Organization settings > Connectors first; you then click Connect.
Free plan will not let you add another connector	The Free plan allows one custom connector. Remove the old one from its three-dot menu, then add MarketIntel.

Symptom	Fix
Claude cannot connect	Confirm the URL ends in /mcp and starts with https. Open <url>/health in a browser. Make sure the server and ngrok are both running and ngrok points to the same port.
No log lines when you add the connector	The request is not reaching your machine. Copy the Forwarding URL from ngrok again (it may have changed after a restart) and edit the connector's URL.
Connected, but Claude does not use the tools	Start a new chat, click + > Connectors and switch MarketIntel on. Mention MarketIntel by name in your prompt.
ngrok says ERR_NGROK_4018 or authentication failed	Add your authtoken: ngrok config add-authtoken <token>. Get the token at dashboard.ngrok.com/get-started/your-authtoken (Step 6).
fastmcp dev server.py gives an error	Use uv run fastmcp dev inspector server.py (needs Node.js), or skip the Inspector and use uv run fastmcp list server.py and fastmcp call.
npx or node not found when starting the Inspector	Install Node.js (brew install node on a Mac) and open a new terminal, or use the fastmcp list / call commands.
uv add fails: Two workspace members are both named marketintel-demo	A pyproject.toml already exists in a parent folder. Delete that one (and its uv.lock and .venv), or create the project in an empty folder, then run uv add again.
Mode shows DEMO but you set a key	Make sure the file is named .env (not .env.txt), is in the same folder as server.py, and has no spaces around the key. Restart the server.
Port 8000 already in use	Change PORT in .env (for example 8010) and run ngrok http 8010.
Tavily errors or empty results	Check the key and remaining credits in the Tavily dashboard. Set FORCE_DEMO_MODE=1 in .env to fall back to sample data.
ModuleNotFoundError	Run the server with uv run server.py from inside the project folder so uv uses the project's packages.

7. Safety and cleanup

- This demo server has no authentication (No sign-in). Anyone who has the ngrok URL can call it, so run it only while demoing and stop ngrok afterwards.
- Claude warns that custom connectors reach unverified services. That is expected here because you built the server yourself; only add servers you trust.
- All tools are read-only. Do not add tools that write or delete data without adding authentication first.
- Keep your Tavily key and ngrok authtoken out of screenshots and out of Git.

- After the demo: press Ctrl+C in the ngrok and server terminals, and remove the connector in Claude (Customize > Connectors, three-dot menu, Remove).